1U Dell PowerEdge R6615 1x KIOXIA CM7 E3.S 7.68TB NVMe

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MLPerf Storage v2.0

Single Host Local Storage

CLOSED

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1. Hardware

The hardware is composed of the client and server being on the same single host. No switch is involved.



1.1 Client

The client is the same 1 Rack Unit Dell PowerEdge R6615 rack server mentioned in section 1.2 below.

1.2 Server

The server is a commercial 1 Rack Unit Dell PowerEdge R6615 rack server providing a single XFS filesystem to the client (itself) (one filesystem total). The configuration is the default one with 1x KIOXIA CM7 E3.S NVMe drive with 7.68 TB capacity. The server communicates with the client through localhost as it is the same system. It contains a single socket AMD EPYC 9454P 48 Core Processor with 12x 32GB DDR5 DIMMS for a total of 384 GB of host memory. Power supply is in active/passive configuration where each power supply is rated at 1400 W.

1.3 Switch

There is no switch involved, as it involves only one rack server utilizing localhost to connect.

2. Software

2.1 Client

The clients distribution is the same as section 2.2 described below, as it is a single rack server.

2.2 Server

The server's distribution is Ubuntu 24.04.2 LTS with kernel 6.8.0-60-generic. The server is using an XFS file system on a single drive to host the training and checkpointing files.

2.3 Switch

There is no switch involved, as it involves only one rack server utilizing localhost to connect.

3. Settings

3.1 Client

Refer to section 3.2 as it is the same rack server.

3.2 Server

No tuning was applied on the Dell PowerEdge R6615 rack server. The local XFS filesystem was mounted on a single drive with noatime and nodiratime mounting flags.

3.3 Switch

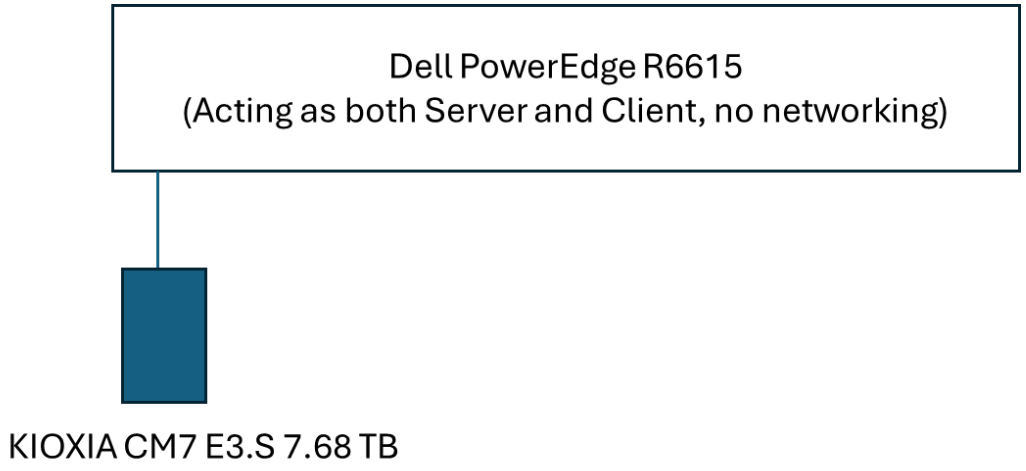
No tuning was applied, as no switch was involved, and everything was done via localhost.

4. Power Requirements

Power Requirements Table

System component	Nameplate rated power	Design power
Dell PowerEdge R6615	1400	1400
KIOXIA CM7 E3.S 7.68TB NVMe	25 watts	25 watt
Total		1425 watt

5. System Topology



6. Misc

6.1 Drop cache

The cache of the Dell PowerEdge R6615 server was dropped between ten iterations of saving write checkpoints and 10 iterations of loading read checkpoints.

The following commands were run to produce the checkpointing results.

mlpstorage checkpointing run --client-host-memory-in-gb 384 --model llama3-8b --num-processes 8 --checkpoint-folder /mnt/mlperf/checkpoint_test --num-checkpoints-write=10 --num-checkpoints-read=0 --closed

echo 3 > sudo tee /proc/sys/vm/drop_caches

mlpstorage checkpointing run --client-host-memory-in-gb 384 --model llama3-8b --num-processes 8 --checkpoint-folder /mnt/mlperf/checkpoint_test --num-checkpoints-write=0 --num-checkpoints-read=10 –closed

6.2 No tunings

No additional tuning was applied to the Dell PowerEdge R6615 rack server.

6.3 Logs

All the logs generated were taken from the mlpstorage utility.